

Contact

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(LinkedIn)
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Top Skills

Strategic Planning
Strategic Consulting
TypeScript

Languages

Hindi (Full Professional)
Bengali (Native or Bilingual)
English (Native or Bilingual)
German (Elementary)

Certifications

IT Information Library Foundations
Certification (ITIL)
React Native Course
Machine Learning in JavaScript with
TensorFlow.js
Introduction to Generative AI
International Software Testing
Qualifications Board Foundation

Publications

TorchEsegeta: Framework for
Interpretability and Explainability of
Image-based Deep Learning Models
An Automated Tongue Tracker for
Quantifying Bulbar Function in ALS

Budhaditya Mukhopadhyay

Senior AI Engineer | Strategic Advisor – UAV & AI Innovation | Edge
AI | ML & DL
Munich, Bavaria, Germany

Summary

Senior AI Engineer specializing in audio processing and edge AI, with a Master's degree in Data Science from Otto von Guericke University. I build and optimize production-grade Machine Learning and Deep Learning systems, with a focus on real-world deployment in healthcare. My expertise includes audio-based AI, noise-robust modeling, and on-device inference, with additional experience in medical imaging and computer vision. Alongside my core role, I serve as a Strategic Advisor at Weevils Drones, supporting AI-driven innovation, UAV intelligence, and international technology partnerships. Previously, I gained 4.5+ years of IT experience, working on enterprise web services using TIBCO technologies.

Experience

VisionHealth GmbH

3 years 3 months

Senior AI Engineer

February 2026 - Present (2 months)

Munich

- Leading end-to-end ML systems for audio-based healthcare applications
- Designing and deploying on-device deep learning models in real-world conditions
- Owning data strategy, model training, validation, and production deployment
- Collaborating cross-functionally to ship clinically reliable AI

AI Engineer

January 2023 - Present (3 years 3 months)

Munich, Bavaria, Germany

- Implementation of AI models using TensorFlow.js for detecting inhalation, spray, exhalation, and proper manoeuvres.
- Integration of AI into audio and video processing pipelines, leveraging React Native for cross-platform mobile development.

- Utilization of temporal modeling techniques to enhance efficiency and accuracy in real-time data processing.

Weevils Drones

Strategic Advisor

May 2025 - Present (11 months)

Owning the end-to-end architecture and production deployment of AI-powered drone systems, bridging hardware, software, and machine learning into scalable, operational platforms.

Implementing AI-based drone tracking and intelligence solutions, ensuring robustness, performance, and real-world deployability.

Mentoring interns, advising leadership on strategy, partnerships, and market positioning, and supporting business development and go-to-market initiatives.

Volkswagen AG

1 year

Data Science Researcher : Master Thesis

April 2022 - December 2022 (9 months)

Wolfsburg, Lower Saxony, Germany

Development of a motion sickness detection solution using deep learning

- Prediction of motion sickness using contextual (tabular) data using ML/DL algorithms followed by explainability
- Finding correlation between pupil diameter to motion sickness inside car using convolution blocks
- Prediction of next state of pupil diameter in car for future prediction using LSTM
- Using computer vision to detect in car motion sickness

Data Science Intern

January 2022 - March 2022 (3 months)

Germany

Development of a motion sickness detection solution using deep learning

- Detection of motion sickness using time series data of pupil diameter by using LSTM and 1D convolution techniques
- Using the features extracted from frames of a video for training an LSTM-CNN block
- Worked on the handling large data of face images and face landmark data points for training model

- Written custom scripts for extraction of data from video files.

Otto-von-Guericke University Magdeburg

1 year 11 months

Student Research Assistant

November 2021 - September 2022 (11 months)

Magdeburg, Saxony-Anhalt, Germany

Working on the development of a one-stop solution for distribution package for python library (PyTorch)

- Downloading, extraction, and creation of custom dataset packages for easy usability of
- Pipeline supporting text, audio, and image packages

Development of a solution for German Speech to text translation using deep learning

- Using python FLASK for interaction between packages and running an online model for translation
- Fine-tuning model for specific tasks
- Implementation is done using CI/CD (Docker)

Student Research Assistant

November 2020 - February 2022 (1 year 4 months)

Magdeburg, Saxony-Anhalt, Germany

Development of a solution for detecting motion and blur in medical scans

- Development of Deep Learning-based python application for detecting blur in MRI and CT scans
- Pipeline supporting supervised training and testing on 2D and 3D volume
- Support for Multi GPU and Multi-threading to enhance training and testing
- Using explainable AI to interpret the reasons for classifying a scan as Blur or Not Blur

German Center for Neurodegenerative Diseases (DZNE)

Research Assistant

August 2020 - October 2021 (1 year 3 months)

Germany

- Created a video processing pipeline using computer vision and Convolution Neural Network(CNN)
- Pipeline can handle the following pre-processing automatically:
 - # Denoising video(to handle camera shakes)
 - # Video standardization - both format and dimensions
 - # Video cropping based on the user's selected time frame

- Created custom dataset of lip section with manual segmentation done on the tongue area
- Increase the number of samples in the dataset by using automatic segmentation(active learning) and augmentation
- Performed face extraction followed by lip tracking in video files using FaceNet
- Bounding box creation around lips to extract the portion for tongue detection using YOLO, RCNN, OpenCV, Dlib, Tensorflow, PyTorch, PIL, Scikit-learn, Scikit-image packages
- Used custom deep neural network based on MobileNet-V2 architecture to perform tongue segmentation in individual frames of the video files
- Model was pre-trained on ImageNet and fine-tuned on custom made tongue dataset for feature extraction
- 3D depth estimation from 2D images using deep learning to get the distance of the person from the camera
- Developed an interactive frontend GUI tool using python FLASK
- Used Seaborn and Matplotlib to visualize the statistical results
- Created a self-sufficient and platform-independent container package of the tool using Docker

Cognizant

Associate

October 2018 - September 2019 (1 year)

Kolkata Area, India

- Development of Web Services (APIs) for Pharmacy Domain
- Development of Automated validation scripts for pre-deployment to production server
- Preparing Functional and Technical Design Documents

Ericsson

Solution Integrator

February 2015 - October 2018 (3 years 9 months)

Kolkata Area, India

- Development of Web Services (APIs) for a Full Stack Telecom Transformation Project
- Worked as a SPOC for TIBCO team at onsite customer location (Kuwait), representing our offshore team

Ericsson

Intern

June 2012 - July 2012 (2 months)

Kolkata

Development of a Server Monitoring tool - worked on Linux Kernel and implemented an Alerting System for different overloads in servers.

Education

Otto-von-Guericke University Magdeburg

Master's degree, Data and Knowledge Engineering · (2019 - July 2023)

West Bengal University of Technology, Kolkata

Bachelor's Degree, Information Technology · (2010 - 2014)